

Operations and financial plan

How the system works, what it costs, what to charge, how to save on routes, which machines to buy and when it starts making money. Málaga, November 2026.

COSTA CLEAN · WORKING DOCUMENT

1. Executive summary

Proposition	The customer leaves a bag, we collect it by bike and return it within 24-48 h washed, dried, ironed, folded, vacuum packed and with the scent they chose.
Where	Málaga city, starting with Centro, Soho, Malagueta, Huelin, Carretera de Cádiz, Cruz de Humilladero, Ciudad Jardín and Pedregalejo (Zone 1). El Palo and Teatinos as Zone 2 with a surcharge.
Initial team	2 riders on electric cargo-bikes + 3-5 part-time laundry staff in a micro-hub + you (operations, customer service, B2B sales).
Initial investment	**≈ €45,000 excl. VAT** (≈ €54,500 incl. VAT) setting up the hub with new machinery. **≈ €20,000-25,000** starting with second-hand machines and a single new cargo-bike.
Selling price	5 kg bag €22.90 · 10 kg €36.90 · 15 kg €49.90 (VAT included, pickup and delivery included). Full ironing + €12/20/28.
Cost per order	€37.70 at 15 orders/day · **€27.80 at 30 orders/day** · €23.85 at 50 orders/day (average order 8 kg).
Break-even	**≈ 18 orders/day (≈ 390/month)** with 2 full-time riders.
Margin	20 % at 30 orders/day (≈ €4,500/month) · 31 % at 50 orders/day (≈ €11,900/month).
Main risk	Hub labour is half the cost. If ironing gets out of control or the laundry staff are not hired properly, the business does not add up.

2. How the system works (operations)

2.1 Order flow

Customer orders (web/WhatsApp) → Awaiting pickup
↓ rider collects in a 2 h slot, weighs the bag in front of the customer
Collected (on the bike) → At the hub (weighed, sorted, QR-labelled, scent noted)
↓ laundry staff assign it to a free washer (app /hub)
Washing (45 min) → Drying (40 min) → Ironing → Vacuum packing → Ready
↓ the optimiser groups ready deliveries by zone and assigns them to a rider
Out for delivery → Delivered (photo + confirmation in the rider app)

Every status is logged with time and person in the app, so at any moment you know **what is being washed, what is sitting idle waiting for a machine, what is ready and where it has to be delivered**, and the customer can receive an automatic WhatsApp at each step.

2.2 Where the washing happens: at the staff's homes or in a hub?

Your starting idea is that several local women run the washing machines and do the packing. There are two ways to do this and it is worth deciding before buying anything:

	A. HOME NETWORK (EACH PERSON WASHES AT HOME)	B. MICRO-HUB (PREMISES WITH MACHINES, STAFF WORK IN SHIFTS)
Investment	Minimal (bags, small vacuum sealers, scents)	€25,000-45,000 in machines and installation
Capacity	Domestic 8 kg washer, cycle + drying ≈ 2.5-3 h → 2-3 orders/day per person	Professional 11-20 kg washer, 45 min cycle → 8-10 orders/day per machine
Energy cost	Domestic dryer: very inefficient, and the worker pays for it (legal and motivation problem)	Professional dryers, industrial tariff
Quality and traceability	Hard to standardise (household smells, pets, customers' loads mixed up)	Uniform process, scent control, scales, labels
Vacuum packing and ironing	Slow domestic sealer; domestic iron	Chamber vacuum sealer, steam ironing table with boiler (3x faster)
Legal	**Home-based work = employment relationship**. Paying someone "per load" without a contract is a bogus self-employed arrangement: fine + back contributions.	Normal part-time contracts (dry-cleaning and laundry collective agreement)
Routes	The rider has to visit N homes on top of the customers: more km	A single loading and unloading point

Recommendation: micro-hub (B), with the same women as part-time employees. It is more investment, but it is the only way to offer uniform ironing and vacuum packing and to scale to 30-50 orders/day. The home network (A) only makes sense as a 1-2 month proof of concept with 10-15 trusted customers to validate demand before signing a lease; even then the people must be registered as employees, however few hours they work.

2.3 The hub

- Premises of **100-120 m²** with three-phase power (or the option to install it), drains and an air outlet for the dryers. Ideal location: **Perchel / Cruz de Humilladero / around María Zambrano station**: 10-15 minutes by bike from every Zone 1 neighbourhood and rents are lower than in the Centro. Alternative: the San Rafael or Guadalhorce industrial estates (cheaper, but 20-25 min by bike to the Centro). Indicative rent: **€900-1,500/month + VAT** (2025-26 listings on Fotocasa / CompanySpace).
- Areas: reception and weighing · 3 washers · 3 dryers · 2 ironing stations · folding and packing table · shelving for ready orders sorted by route.
- Shifts: 7:00-15:00 and 15:00-21:00. Afternoon pickups are washed first thing the next morning; morning pickups are delivered in the afternoon (24 h) or the next day (48 h).

2.4 Bike delivery

- **2 electric cargo-bikes with a box** (capacity 80-100 kg / 4-6 bags of 10 kg). Málaga city is flat along the whole coastal strip; only Ciudad Jardín and Teatinos have gentle hills.
- Each rider makes **≈ 7 stops/hour** in dense areas (4 min per stop + 3-4 min pedalling). With 7 hours on the road: **≈ 50 stops/day per rider = 25 orders** (each order is one pickup + one delivery). Capacity of the pair: **≈ 50 orders/day**. Above that, a third rider is needed.
- The rider's phone sends its GPS position to the app every 5 s: you see it on the map and the customer can get a "your rider is 5 minutes away" message.

2.5 Scents and vacuum packing (the differentiator)

- 6 scents: sea breeze, lavender, clean cotton, Mediterranean citrus, Málaga jasmine and unscented (hypoallergenic). Applied with **professional fabric perfume in the final rinse** (≈ €0.03/kg) and a final spray before packing. Loads must be washed by scent, or washers with automatic dosing used.
- Vacuum packing: 60-70 litre PA/PE bags (€0.50-0.70 each), 2 per 8 kg order. A **chamber vacuum sealer** (€1,500-3,500) takes 30-40 s per bag; external sealers (€300-600) work but take 2-3 min and the embossed bags cost twice as much. Vacuum compresses the clothes by ≈ 60 %: more orders fit in the bike box and the customer perceives "new clothes".
- Careful with ironing: a fully compressed vacuum pack creases ironed clothes. Use a **partial vacuum** (leave 20-30 % of the air in) and fold with card or tissue paper between shirts. Test it on your own clothes before selling it.

3. Assumptions of the financial model

ASSUMPTION	VALUE	SOURCE
Average order	8 kg → M bag (10 kg) at €36.90 incl. VAT	Expected mix 30 % S, 50 % M, 20 % L
Orders with full ironing	25 % (+€20 on average)	Estimate
Operating days	22/month (Monday-Saturday would be 26; prudence)	
Employer cost of laundry staff	**€12.50/h**	SMI 2026 (minimum wage): €1,221/month × 14 payments = €17,094/year; with ≈ 31.5 % employer social security ≈ €22,500/year ≈ €1,875/month ≈ €11.60/h; €12.50/h leaves room for the collective-agreement supplement
Employer cost of a full-time rider	**€1,900/month**	Same SMI 2026 calculation
Labour time per 8 kg order	**54 min** (basic ironing) / **114 min** (full ironing)	Sort and load 8 · move to dryer 5 · fold 18 (≈ 35 garments × 30 s) · basic ironing 15 (6-8 garments) · pack 8 · full ironing +60
Consumables per order	**€3.54**	Detergent + softener + perfume €0.12/kg → 0.96 · 2 vacuum bags 1.20 · 6.7 kWh × €0.18/kWh = 1.21 · 80 L of water × €2.20/m³ = 0.18
Electricity	€0.18/kWh	SMEs on indexed tariffs in 2025-26 pay between €0.13 and €0.20/kWh depending on contracted power
Rent	€1,100/month	See 2.3
Cargo-bikes	€320/month for both	€4,000 each amortised over 36 months (≈ €110) + maintenance €35 + insurance €15
Software, insurance, accountant, fixed utilities, marketing	150 + 100 + 150 + 150 + 400 = **€950/month**	
Machinery depreciation	€30,000 over 7 years = **€357/month**	See section 7

4. What does it cost us? Monthly and per-order costs

ORDERS/DAY	15	30	50
Orders/month · kg/month	330 · 2,640	**660 · 5,280**	1,100 · 8,800
Hub labour	€4,744	**€9,488**	€15,813
Riders (2 full-time)	€3,800	**€3,800**	€3,800
Consumables and energy	€1,169	**€2,338**	€3,896
Fixed costs (rent, bikes, depreciation, software, insurance, accountant, marketing)	€2,727	**€2,727**	€2,727
Total cost/month	**€12,440**	**€18,352**	**€26,236**
Cost per order (8 kg)	**€37.70**	**€27.81**	**€23.85**
Cost per kg	€4.71	**€3.48**	€2.98
Rider utilisation	31 %	**61 %**	102 %
Part-time laundry staff needed	4.3	**8.6**	14.4

Cost per order breakdown at 30 orders/day:

ITEM	€/ORDER	%
Hub labour (69 min on average)	14.38	52 %
Delivery (pickup + drop-off)	5.76	21 %
Consumables, energy and water	3.54	13 %
Allocated fixed costs	4.13	15 %
Total	**27.81**	

Key takeaways:

1. **Hub labour rules everything.** €14.40 of every order is laundry staff hours. Every 5 minutes shaved off the process (standardised folding, boiler iron, chamber vacuum sealer) saves ≈ €1/order = €660/month at 30 orders/day. That is why a home network with slow machines does not add up.
2. **Riders are an almost fixed cost.** At 15 orders/day they are 31 % utilised: starting with **1 full-time rider + 1 part-time** saves €950/month until you pass 25 orders/day.
3. **The kg mix matters little for cost, a lot for revenue.** Washing 15 kg costs only €2-3 more than washing 8 kg (almost the same labour), but it brings in €13 more. Push the L bag and bed-linen packs.

5. What price do we sell at?

5.1 What the competition charges

SERVICE	WHAT IT INCLUDES	PRICE
Mr Jeff (national, also in Málaga)	30° wash, dry and fold. Free pickup and delivery. No ironing.	5 kg €13 · 8 kg €17 · 15 kg €24 · subscription 4 bags/month €60-64
Washrocks (national)	Wash and fold	5 kg €20.95
Lavandería Industrial Málaga / Bago	Per kilo, business-oriented, with delivery	€1.50-1.80/kg
Self-service laundrettes in Málaga	Machine only	€3-5.50 per 8-10 kg wash

Conclusion: **you cannot compete on €/kg with basic washing** (Mr Jeff sells 8 kg for €17, and our cost at 30 orders/day is €27.80). Costa Clean is a different product: ironing, vacuum, scent, bike delivery in a 2 h slot. The target customer is the one who today pays €20-25 for an hourly ironing service or takes shirts to the dry cleaner (€3-4 per shirt).

5.2 Recommended price list (VAT included)

PRODUCT	PRICE	€/KG	WHAT IT INCLUDES
S bag · up to 5 kg	**€22.90**	4.58	Zone 1 pickup and delivery, scent, basic ironing, vacuum packing
M bag · up to 10 kg	**€36.90**	3.69	Same
L bag · up to 15 kg	**€49.90**	3.33	Same
Full ironing	+€12 / +€20 / +€28		Every garment ironed (S/M/L)
Zone 2 surcharge (El Palo, Teatinos)	+€2.90		
"Always clean" subscription	€129/month		4 M bags a month (€32.25/bag, -13 %), fixed day, simultaneous pickup and delivery
Duvet / comforter	€19.90		Washed, dried, vacuum packed (sells on its own; vacuum is a huge value here)
B2B holiday rental	**from €14.90 + VAT**		Bed + bath kit (≈ 4 kg: sheet set, 2 towels, 2 hand towels), calender-ironed or folded, delivered by check-out

With these prices, at 30 orders/day:

	€
Average price charged (M bag + 25 % with full ironing)	€41.90 incl. VAT
Net revenue excl. VAT	€34.63
Cost per order	€27.81

	€
Margin per order	**€6.82 (20 %)**
Minimum price for a 30 % gross margin	€48.07 incl. VAT

The margin is tight until you pass 40 orders/day. Levers to improve it without raising list prices: subscriptions (recurring revenue and predictable routes), **B2B holiday rentals** (Málaga has more than 10,000 tourist apartments; bed linen is fast to process and fills the machines in off-peak hours), and selling full ironing and duvets as high-margin extras.

5.3 Break-even

- Contribution per order (net revenue − variable labour − consumables) ≈ **€16.70**.
- Monthly fixed costs (riders + premises + the rest) ≈ **€6,527**.
- **Break-even ≈ 391 orders/month ≈ 18 orders/day.**
- With 1.5 riders instead of 2: ≈ 334 orders/month ≈ 15 orders/day.

A realistic target: 10-15 orders/day in month 3, 25-30 in months 6-9, 50 in year 2 with a third rider and one more shift at the hub.

6. How we save on routes

6.1 Golden rules

1. **Zone by day.** Zone 1 West (Huelin, Carretera de Cádiz, Cruz de Humilladero) Mondays and Thursdays; Centro/Soho/Malagueta every day (that is where the density is); Pedregalejo/El Palo Tuesdays and Fridays; Teatinos/Ciudad Jardín Wednesdays and Saturdays. The customer picks a day among those available for their neighbourhood. This multiplies stop density per route, which is what decides delivery cost.
2. **2-hour slots, never "whenever you like".** It lets you chain 10-14 stops per slot.
3. **Pickup and delivery in the same visit** for subscribers: the rider drops the clean bag and takes the dirty one. **One stop instead of two**: it cuts delivery cost per order from €5.76 to €2.88.
4. **The hub as the single origin.** The rider leaves loaded with the zone's deliveries, picks up along the same streets and comes back with the collections. The app orders the stops with a nearest-neighbour algorithm improved with 2-opt (on a 15-20 stop route it saves 15-30 % of km versus the order in which orders arrived).
5. **Weight in the box.** With vacuum, 6 M bags fit in a cargo-bike box. Without it, 3-4.

6.2 Numbers

	ELECTRIC CARGO-BIKE	SMALL VAN
Investment	€4,000	€18,000-25,000 (or leasing €350-450/month)
Cost/month (depreciation, energy, maintenance, insurance)	**€160**	€700-900 (leasing + diesel for 1,800 km + insurance + parking)
Stops/hour in the Centro	**7** (parks at the door)	4-5 (finding parking, blue zone, pedestrian Centro)
Km/day per vehicle	35-45	50-70
Licence, fines, low-emission zone	No	Yes (Málaga has a Low Emission Zone in the Centro)

Two cargo-bikes versus one van: **saving ≈ €500-700/month** and, above all, 40 % more stops per hour in the Centro, which is where the customer is. A van only makes sense for the holiday-rental B2B business (high volume, check-out times) from the second year onwards.

6.3 What the app does

- `/mapa`: optimised route per rider with estimated real km, "unoptimised vs optimised" comparison, live GPS position and the day's track.
- `/rider`: the same route on the phone, navigation button (Google Maps in bike mode), mark as collected/delivered, automatic GPS.
- Still to implement: time slots as a hard constraint and automatic splitting of stops between the two riders (currently assigned from the panel).

7. How much do industrial washing machines cost?

Prices from Spanish online shops 2025-26, **excl. VAT**. Girbau (the Catalan manufacturer, the most widespread in professional laundries) does not publish prices: they are sold on quotation through dealers and usually sit above Primer and Fagor.

7.1 Price references

MACHINE	REFERENCE MODELS	NEW PRICE (EXCL. VAT)
Professional washer 10-11 kg	Primer LP-10 · Fagor LAP-10	**€3,800-3,950**
Professional washer 18-20 kg	Primer RS-18 (Factor 200) · Primer LS-18 (high spin)	**€6,400** on offer (list €8,000) · €10,300 high spin
Washer 15-25 kg (general sector range)	various brands	€2,500-5,000 for basic units, up to €10,000 with high spin
Professional dryer 12-14 kg	Fagor · Primer	**€2,230-2,705**
Dryer 31 kg	Primer DE-28	€7,300
Girbau HS-6013 (13 kg) / ED340 (dryer 13-19 kg)	Girbau	On quotation
Second-hand (Wallapop, laundry closures)	Girbau, Primer, Miele Professional	40-60 % below new; check hours and spin speed

7.2 Recommended configuration for the hub (30-50 orders/day)

ITEM	QTY	€/UNIT	TOTAL
Professional washers 10-11 kg (Primer LP-10 / Fagor LAP-10)	2	3,900	7,800
Washer 18-20 kg for bed linen and B2B (Primer RS-18)	1	6,400	6,400
Electric dryers 12-14 kg (Fagor / Primer)	3	2,700	8,100
Professional ironing tables with boiler and iron	2	900	1,800
Chamber vacuum sealer for textiles + initial consumables	1	2,500	2,500
Three-phase electrical installation, drains, dryer extraction	1	4,500	4,500
Furniture, shelving, scales, trolleys, label printer	1	1,500	1,500
Subtotal machinery and installation			**€32,600**
Electric cargo-bikes with box	2	4,000	8,000
Deposit on premises (2 months) + activity licence and technical project	1	4,500	4,500
Total initial investment excl. VAT			**≈ €45,000**
Total incl. VAT (21 %, recoverable on the tax return)			≈ €54,500

Theoretical capacity of this configuration: 3 washers × 8 cycles/day × 10-14 kg ≈ **300 kg/day ≈ 35-40 orders/day** in one 8 h shift, double with two shifts. It is the right configuration for the first-year target; the dryers are the

bottleneck (drying takes almost as long as washing and you need at least as many dryers as washers).

7.3 How to make it cheaper

- **Minimum start (≈ €20,000-25,000):** 2 second-hand washers of 10-13 kg (≈ €1,800-2,500), 2 second-hand dryers (≈ €1,200-1,500), external vacuum sealer, 1 new cargo-bike + 1 regular e-bike with a trailer, and grow from cash flow. Risk: breakdowns and less capacity.
- **Machinery renting / leasing:** Girbau, Primer and Fagor dealers offer 60-month renting at **≈ 1.8-2.2 % of the value per month** (€30,000 → €550-650/month) with maintenance included. It turns €30,000 of capex into a fixed cost that enters the model with little impact (we already count €357/month of depreciation).
- **Heat-pump dryers:** cost ≈ €1,000 more each but use 40-50 % less energy and need no air outlet. At 30 orders/day they save ≈ €250/month in electricity: they pay for themselves in a year.
- **Gas instead of electricity for the dryers:** cheaper to run, but needs a gas installation and inspections. Only if the premises already have gas.
- **Grants:** Andalucía Emprende and the ICO/ENISA lines for young entrepreneurs; Málaga City Council (Promálaga) has start-up grants. Book an appointment before buying.

8. Risks and how we cover them

RISK	MITIGATION
Damaged or lost garments	Photo of every bag on receipt, QR label per bag, liability insurance covering garments (≈ €100/month), clear terms (we do not wash anything labelled "dry clean only").
Laundry staff: employment relationship	Part-time contracts from day one, Málaga dry-cleaning and laundry collective agreement, occupational risk prevention. Paying "per load" without a contract is the fast track to an inspection.
Ironing eating the margin	Basic ironing as standard (only what needs it), full ironing always as a paid extra, boiler tables, time the work and post times per staff member in the hub.
Low order density at the start	Zone by day, do not open Zone 2 until Zone 1 has 15 orders/day, start with 1.5 riders.
Price competition (Mr Jeff)	Do not compete on basic washing. Always communicate "ironed, vacuum packed and scented".
Vacuum creasing ironed clothes	Partial vacuum, tests before launch, card between shirts.
Activity licence	Laundry activity (IAE 971.1) usually requires a licence with a technical project, soundproofing and ventilation. Confirm with the City Council before signing a lease; avoid premises under homes if possible.

9. Next steps (90 days)

1. **Weeks 1-2:** Validate demand: landing page (`/pedido`) + Instagram + 200 flyers in Soho/Malagueta; target 30 pre-bookings. Wash 10 orders from friends "by hand" to time the real process and test the vacuum with ironed clothes.
2. **Weeks 3-4:** Request 3 machinery quotes (Girbau Málaga, Primer dealer, Fagor Industrial), new and renting, and visit 5 premises in Perchel / Cruz de Humilladero. Check the licence.
3. **Month 2:** Incorporate an SL (€3,000 share capital), register under IAE 971.1, liability insurance, hire 2 part-time laundry staff and 1 rider; buy/rent machinery; install.
4. **Month 3:** Launch in Zone 1 targeting 10 orders/day; sign 3-5 holiday-rental managers to fill the machines in the morning.
5. **Months 4-6:** Second rider, subscriptions, open Zone 2, get hub times below 50 min.

10. Sources consulted

- Professional washer prices (Primer LP-10, Fagor LAP-10, Primer RS-18 / LS-18): Hostelearte, Hosteleria10, Donbar, Gavinox guide, second-hand on Wallapop.
- Dryers: Hosteleria10 catalogue, Primer DE-28 at Donbar, Girbau ED series.
- Competition: Mr Jeff prices, Mr Jeff Murcia (CleanCloud), Mr Jeff Bilbao, Washrocks 5 kg, Lavandería Industrial Málaga, self-service laundrettes in Málaga.
- Labour costs: SMI 2026 and contributions (Cuatrecasas), SMI 2026 guide (Copilot Gestoría), total employer cost (Billeo).
- Premises in Málaga: Fotocasa industrial units Málaga, CompanySpace Málaga.
- Cargo-bikes: Hitway BK6S Plus, Spanish bike sector 2025.
- Water and electricity: Emasa 2024 tariffs (Málaga City Council), electricity prices Spain.